

CANDOR

Anonymous Q&A Platform

Product Case Study

Ashutosh Mohanty

Product Lead

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Executive Summary

Candor is an anonymous Q&A web application: a user shares a public link, anyone who visits it can send an anonymous message, and only the account owner can read the inbox and choose whether to reply publicly, ignore, or report each message.

The product was scoped, built, secured, and deployed to a live production URL in approximately five to six hours, using AI-assisted tools throughout the workflow: Figma Make for initial UI generation, Claude for backend architecture, security review, and deployment debugging, and Replit as the AI-assisted development environment.

This case study documents the product decisions behind Candor, with particular attention to two things that separate it from category incumbents: a genuine access-control model built around a private credential rather than a public username, and a deliberate refusal to build the fake-engagement and paywalled-reveal mechanics that drew FTC enforcement action against both NGL and Sendit, the two products that most directly inspired this build.

Background

This project was built as a hands-on portfolio piece to demonstrate product and technical execution ability for Product Manager, Product Owner, and Business Analyst roles. A prior portfolio project (an IBM Telco customer churn analysis using MySQL and Power BI) demonstrated analytical and BI skills. Candor was deliberately chosen as a different kind of proof point: not a retrospective analysis of a static dataset, but a live, working consumer product that a stranger can open right now and actually use.

The category — anonymous social Q&A — was chosen because it is simple enough to build end-to-end in a short window, but rich enough in real product tension (growth mechanics, safety, monetization ethics, trust) to generate genuine product decisions worth documenting, rather than a superficial CRUD app.

Development Approach

This product was built using AI-assisted software development, not hand-written line by line in the traditional sense. This is disclosed here plainly and is reflected accurately throughout this document. My direct contribution to this project consisted of:

- Product ideation and category selection
- Defining MVP scope and what to explicitly exclude
- Product direction and feature prioritization
- UI iteration and design review
- Directing AI-assisted implementation (frontend, backend, database, security)
- Deployment coordination across three platforms (Replit, Aiven, Render)
- GitHub repository management and version control
- Environment and secrets configuration
- End-to-end manual testing and QA
- Documentation and this portfolio case study

Where this document discusses a technical decision, it reflects a decision I directed and reviewed, executed with AI assistance — not code I hand-typed independently.

Problem Statement

People frequently want candid feedback or engagement from their existing social circle that they won't get through normal, attributed social platforms — the social cost of an honest opinion, attached to your name, is usually too high. Anonymity solves that specific problem, which is why anonymous Q&A apps have repeatedly reached the top of app store charts.

But the two products that popularized this category at scale, NGL and Sendit, solved the anonymity problem by introducing a worse one: both were found by the FTC to have manufactured fake anonymous messages to drive anxious curiosity, then charged a recurring weekly fee for a "reveal" feature that frequently did not work. The core problem Candor addresses is narrower and more honest: can the same anonymous-Q&A mechanic exist without manufactured engagement or a monetized false promise of de-anonymization?

Market Context

Anonymous messaging is a recurring, high-demand category, not a niche one. NGL launched in November 2021 and became the most downloaded app in the U.S. App Store by June 2022, reporting more than 200 million users on its own site. Sendit captured 3.5 million installs within two months of Snapchat banning two predecessor apps, YOLO and LMK, in 2021 — demonstrating that demand in this category outlives any single app's removal.

Both category leaders have since faced federal regulatory action. In July 2024, the FTC reached a \$5 million settlement with NGL and permanently banned it from serving users under 18, after finding the company had generated more than 1,000 fake messages to drive its paid "Pro" subscription. In 2025, the FTC filed a complaint against Sendit alleging approximately 279 million fabricated anonymous messages were sent to drive its \$8.99-9.99/week "Diamond Membership," alongside COPPA violations affecting more than 116,000 children under 13.

This is the market Candor enters as a portfolio exercise, not a commercial product — but the regulatory history of the category is directly relevant product context, and it shaped two of the decisions documented later in this case study.

Target Users and Personas

Three personas represent the distinct behaviors the product needs to support, based on how this category of app is actually used:

Priya — The Casual Sharer

20, college student. Shares her Candor link to her Instagram story mostly for light social validation and fun. Low-stakes usage; wants a quick, low-friction way to see what her circle says about her, and checks her inbox occasionally rather than obsessively.

Devika — The Creator

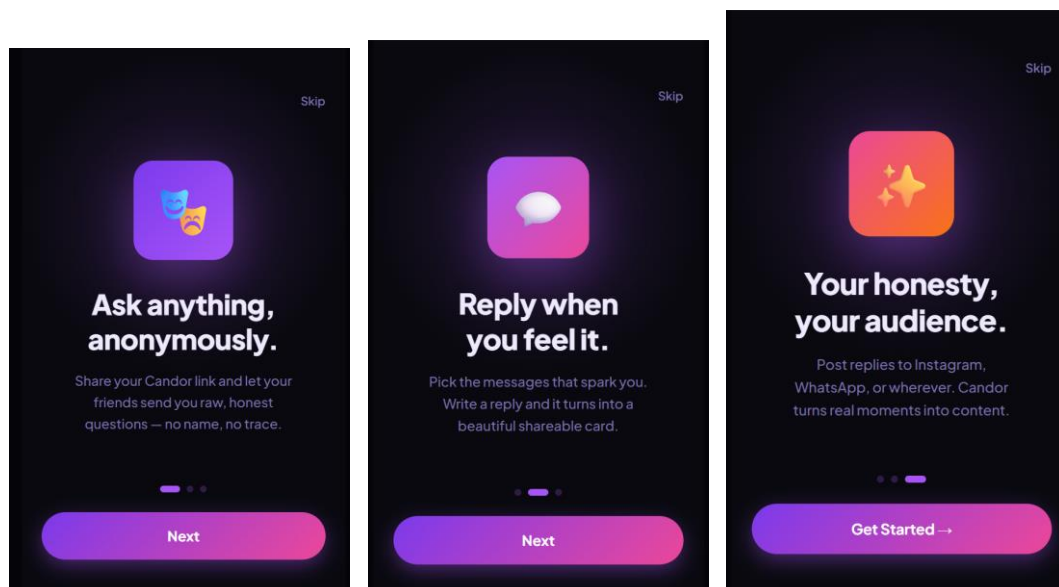
24, micro-influencer. Uses anonymous replies as a recurring content format, turning interesting or funny anonymous messages into public reply cards to keep her feed active. Cares more about the quality of the reply-sharing flow than raw message volume, and is the persona most likely to drive repeat, public exposure of the product.

Rahul — The Curious Sender

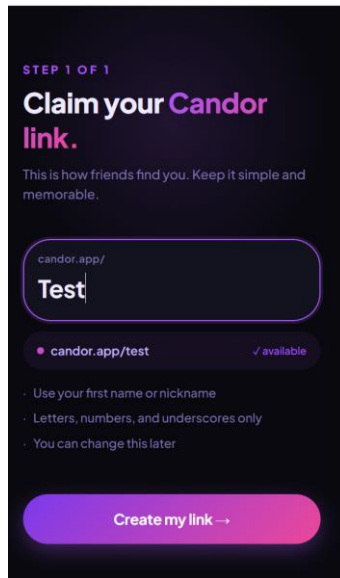
19, rarely promotes his own Candor link but frequently visits friends' links to send messages. His behavior lives entirely on the public, no-login side of the product and is what actually drives the viral loop — a reminder that a meaningful share of a product's user base may never create an account at all.

User Journey

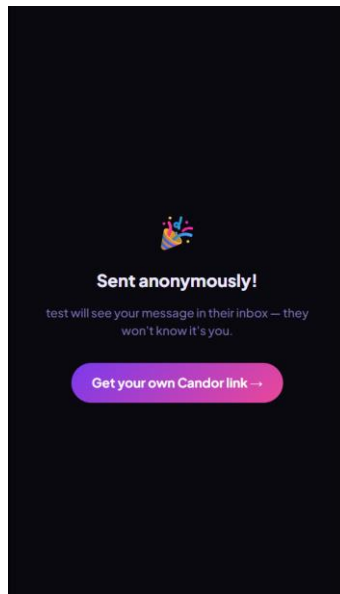
The end-to-end journey spans two distinct experiences: the owner's private app, and the visitor's public, no-login page. A visitor who receives a shared link opens the public send-message page, writes a message, and sends it anonymously with no account required.



Owner onboarding: three screens introducing the core loop before account creation.



Claiming a public username, which becomes the shareable link.



Confirmation shown to an anonymous visitor immediately after sending a message.

On the receiving side, the account owner opens their private inbox, gated by a credential the visitor never sees, and decides whether to reply publicly, ignore, or report each message.

Product Vision and Goals

Vision: a space for honest anonymous conversation between people who already know each other, without manufactured engagement, fake messages, or a paywalled promise of unmasking the sender.

- Ship a fully working, live, independently deployable MVP end to end
- Demonstrate real security thinking, not just feature completeness
- Keep the anonymous-sending side completely frictionless — no account required to send
- Deliberately avoid the manipulative growth and monetization patterns documented in the category's two largest incumbents

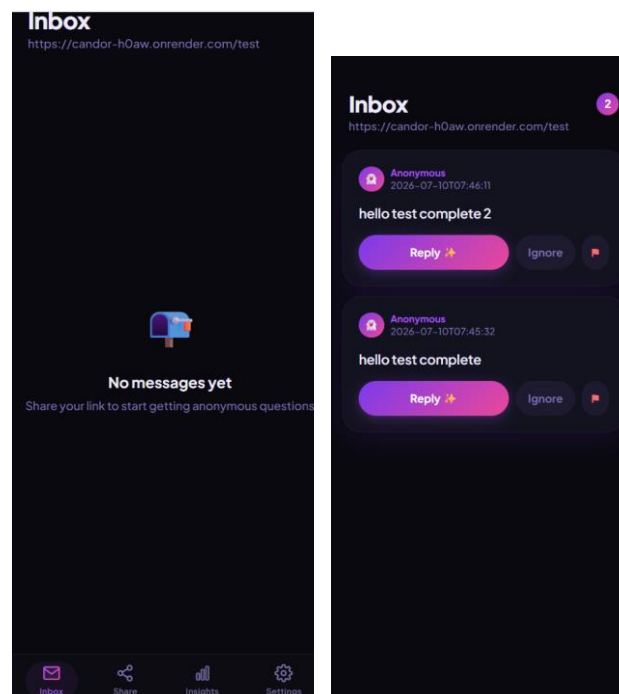
All four goals were met at the time of writing, with the caveats documented in Challenges and Trade-offs and Future Roadmap below.

Success Metrics

Candor has not been publicly launched to a real user base, so this section does not report invented traction numbers. Two things are reported instead: what was actually validated during QA, and the metrics framework that would be tracked if the product were launched publicly.

What was actually validated

The full loop was manually tested end to end on the live production URL: an anonymous message was sent via the public link, appeared correctly in the owner's private inbox, was replied to successfully, and the ignore and report actions both functioned correctly. Access control was explicitly verified — requests without a valid owner token were confirmed to receive 401/403 responses rather than silently succeeding.

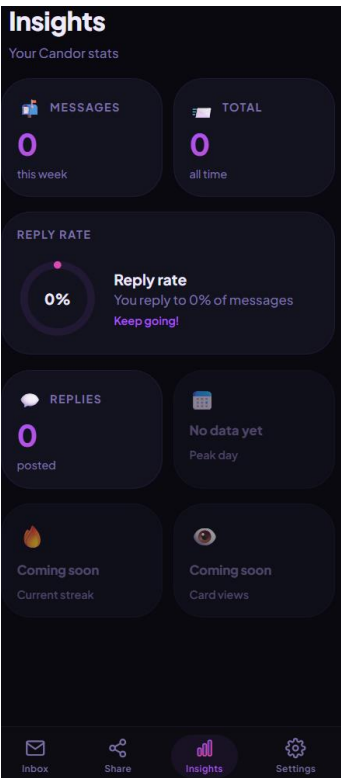


Inbox: empty state, and after receiving anonymous messages.

Metrics framework for a public launch

- Activation rate — percentage of signups who share their link within 24 hours of creating it
- K-factor / viral coefficient — new signups generated per existing user's shared link; the primary growth lever, since virality lives entirely on the public sending side rather than app-store discovery
- DAU/MAU ratio — stickiness of the owner-side inbox-checking habit
- Messages sent per active link per week — volume health of the sending side
- Reply rate — percentage of received messages that get a public reply; already computed in-product on the Insights screen
- D1 / D7 / D30 retention — whether owners return to check their inbox over time

- Report rate — percentage of messages flagged; tracked as a safety signal, not a vanity metric



The in-product Insights screen, showing real reply-rate and activity data rather than placeholder figures.

Competitive Inspiration

NGL and Sendit are the two products that most directly inspired Candor's core mechanic: a public shareable link, anonymous sending, and an owner-side inbox with a public-reply option. The table below compares the three products on the dimensions that mattered most while making product decisions for Candor.

Dimension	NGL	Sendit	Candor
Fabricated messages	FTC found 1,000+ AI-generated fake messages used to drive subscriptions	FTC alleged ~279 million fake anonymous messages sent	None. Every message in the inbox came from a real visitor.
"Reveal sender" feature	Paid "Pro" tier promises hints; FTC called the reveals largely non-functional	Paid "Diamond" tier (\$8.99-9.99/wk); reveals often fabricated per FTC complaint	Not built, on principle — anonymity isn't sold back to the sender's target
Access control	Not publicly documented	Not publicly documented	Private owner token (separate from public username) required for every inbox action
Regulatory outcome	\$5M FTC settlement; banned from serving under-18 users (Jul 2024)	Active FTC complaint over COPPA violations (2025)	N/A — portfolio project, not a commercial product

MVP Definition and Feature Prioritization

The MVP boundary was set deliberately narrow, with one category of feature excluded on principle rather than simply deferred:

Priority	Scope
Must	Public link generation; anonymous send (no login); private inbox; reply-as-public-card; ignore/report; owner-token access control
Should	Share screen with real link + QR code; basic in-product insights (message volume, reply rate, most active day)
Could	Rate limiting on the public send endpoint; branded share icons; favicon
Won't (v1)	Paywalled "reveal sender" feature (excluded by design, not deferred); push notifications; automated content moderation beyond manual report; native mobile app

Product Decisions

Username-based public links, not phone or email. This minimizes friction for anonymous senders, who never need to create an account. The direct consequence — a username is public by definition and therefore cannot double as proof of identity — is what motivated the owner-token access-control model documented in Security Decisions.

"Ignore" is a hard delete, not an archive. This matches the mental model of dismissing an anonymous message rather than filing it away. The trade-off — no undo — is treated here as a documented product choice, not an oversight.

No "reveal sender" feature of any kind, paid or free. This was excluded on principle, not simply left out of MVP scope. Both NGL and Sendit sold a reveal feature that regulators found to be largely non-functional; building a version of that feature, even an honest one, would reintroduce the exact trust problem this project set out to avoid.

Technical Overview

Frontend: React 19 with TypeScript, built with Vite and styled with Tailwind CSS. React Router separates the owner's authenticated app (onboarding, inbox, reply, share, insights, settings) from the public, unauthenticated route that any visitor can open to send a message.

Backend: a Python Flask REST API, served in production by Gunicorn. The database is MySQL, hosted on Aiven, with three normalized tables — users, messages, and replies — connected by foreign keys.

Deployment: the application is containerized with Docker and deployed on Render, which auto-deploys on every push to the GitHub repository's main branch.

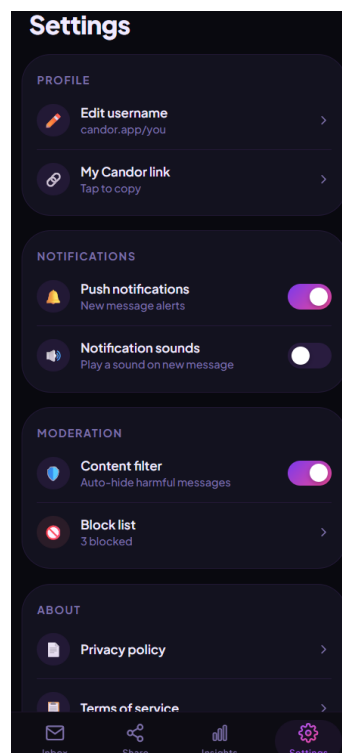
Security Decisions

The most consequential technical decision in this project was recognizing that a public username cannot serve as an authentication credential. Early in development, every inbox action — reading messages,

replying, ignoring, reporting — was reachable by anyone who knew a username or a message ID, since the only "identity" check was the username itself, and the username is, by design, the exact thing printed on the public link.

The fix was a private `owner_token`, generated with Python's `secrets` module at signup and never derivable from the username. Every sensitive backend route now requires this token via a request header, verifies that the token's owner actually matches the resource being accessed or modified, and returns a 401 or 403 response otherwise. This was verified directly: requests without a valid token were confirmed to be rejected rather than silently succeeding.

A rate limit (10 requests per minute per IP address) was added to the public, unauthenticated send-message endpoint specifically, to reduce the risk of spam or abuse on the one route that intentionally has no login requirement.



Settings screen: content filtering and block-list controls available to the account owner.

Deployment Journey

The frontend UI was initially generated using Figma Make, covering the account owner's experience — onboarding, signup, inbox, reply, share, insights, and settings. Figma Make did not generate the public, no-login page a visitor uses to actually send a message, which was identified as a gap and built separately, since it is the mechanic the entire product depends on.

The backend, MySQL schema, and initial deployment were set up on Replit, connected to a free-tier MySQL instance on Aiven. Moving from Replit's own hosting to a dedicated free-tier deployment on Render surfaced three real defects, each diagnosed and fixed in turn:

- The Docker build failed with "vite: not found." The project's lockfile indicated pnpm, but the Dockerfile was installing dependencies with npm, which silently ignored the pnpm lockfile and, combined with Docker's default production environment, skipped installing devDependencies —

including the build tool itself. The first attempted fix (forcing `NODE_ENV=development` during install) did not address the actual cause and failed identically; the real fix was installing and building with `pnpm`.

- Visiting a real, existing user's public link returned a raw 404 instead of the anonymous-send page. Flask's built-in static file handler was registered at the application root and took priority over the catch-all route meant to hand unmatched paths to the React frontend, so any path that didn't match a literal file on disk failed before the frontend ever loaded. The fix removed Flask's built-in static handler entirely and let the catch-all route own all non-API traffic.
- The Share screen displayed a hardcoded placeholder domain left over from the Figma Make export, rather than the application's real live URL. This was only caught through manual QA — clicking the actual link the app displayed to users.

Challenges and Trade-offs

Two challenges are worth documenting in more detail than the deployment log above, because they reflect judgment calls rather than pure debugging:

Root-cause discipline under time pressure. The first fix attempted for the Docker build failure (adjusting an environment variable) was a reasonable hypothesis that turned out to be treating a symptom. Confirming the actual root cause — a package-manager mismatch — required a second, more skeptical debugging pass rather than accepting the first plausible explanation.

Recognizing an incomplete security fix before shipping it. An initial plan to fix the access-control gap by checking "does this username own this message" was reconsidered before implementation, once it was clear that the username is the public link itself and therefore proves nothing about identity. This produced the private owner_token design instead. Documenting this here deliberately, rather than presenting only the final design, because the willingness to discard a plausible-but-insufficient fix is itself a product and engineering judgment worth showing.

Free-tier hosting as an explicit constraint. Both Render's free web-service tier and Aiven's free MySQL tier were chosen deliberately over paid alternatives. The trade-off is a cold-start delay of roughly 30-50 seconds after 15 minutes of inactivity — acceptable for a portfolio project reviewed by a recruiter, but flagged here explicitly as something a real commercial launch would need to revisit.

Lessons Learned

- A public, shareable identifier (a username, a link) can never double as an authentication credential — this should be caught at design time, not discovered later during a security review.
- A plausible first fix for an infrastructure failure is not the same as the actual root cause; it's worth pushing one level deeper before considering a bug closed, especially under time pressure.
- AI-assisted development meaningfully compresses build time — this entire product, from empty repository to a live, secured, publicly usable URL, took roughly five to six hours — but it does not remove the need for active product judgment. The two most important gaps in this build (the missing public send screen, and the initial insufficient access-control design) were both caught by directed review, not by the tooling itself.

Future Roadmap

- Automated unit and integration tests (currently validated through manual end-to-end QA only)
- Push notifications for new messages
- Expanded automated content moderation, beyond the current manual report flag
- A custom domain, replacing the current Render-provided subdomain
- A native mobile wrapper, if usage patterns justified it

Conclusion

Candor demonstrates a complete product cycle compressed into a single short build window: a real problem statement grounded in a well-documented category, explicit product decisions with visible trade-offs, a live and independently verifiable deployment, and a security fix that was revised once before being considered correct rather than shipped on the first plausible attempt. The product is live and testable at candor-h0aw.onrender.com, with source available at <https://github.com/AshutoshMohanty07/Candor>.